Test on One Dimensional Array (Except Algorithms)

Marks: 70

Multiple choice questions: **20 x 1**

1. Array is also known as
   a. Subscripted Variable b. Indexed Variable
   c. Linear Variable d. None

2. Array is ….(Multiple Answers)
   a. Built-in data type b. User defined Data Type
   c. Primitive Data Type d. Composite Data Type

3. Size of array
   a. Can be modified b. Can't be modified
   c. Decreased only d. Increased only

4. double a[]= {3.5, 2.3, 1.4, 5.0, 9.2, 4.8}; physical size of this array is
   a. 24 bytes b. 48 bytes
   c. 96 bytes d. 12 bytes

5. An array A with base address 200, of data type int, physical address of A[5] would be
   a. 224 b. 216
   c. 220 d. 228

6. Maximum number of exchanges possible in bubble sort applied on an array with size 10
   a. 55 b. 45
   c. 35 d. 66

7. Which of the following is TRUE about arrays in Java?
   a. Java arrays are passed by reference to methods.
   b. Java arrays can hold elements of different types.
   c. Array index in Java starts from 1.
   d. Java automatically resizes arrays when needed.

8. Which of the following correctly creates an array of 5 integers and initializes all values to 0?
   a. int arr[5] = new int[]; b. int[] arr = new int[5];
   c. int arr = int[5]; d. int arr[] = new int(5);

9. What is the default value of an uninitialized element in an integer array in Java?
   a. 0 b. null
   c. Garbage Value d. It depends on compiler

10. Which of the following is **not** a valid way to declare an array in Java?
    a. int[] a = new int[5]; b. int a[] = new int[5];
    c. int a[5]; d. int[] a; a = new int[5];

11. Which of the following would throw an ArrayIndexOutOfBoundsException?
    a. Accessing arr[-1] b. Accessing arr[arr.length - 1]
    c. Accessing arr[0] in a 5-element array d. Initializing an array with size 0

12. Consider: int[] a = new int[3];. How many reference variables and how many objects are created?
    a. 1 reference, 1 object b. 3 references, 1 object
    c. 1 reference, 3 objects d. 3 references, 3 objects

13. What does arr[arr.length] do?
 a. Returns the last element b. Throws ArrayIndexOutOfBoundsException
 c. Compiles but returns garbage value d. Resets the array

14. int a[]= {4, 5, 6, 7, 2, 1}; int s= a[1] + a[2] + a[1+2]; s= ?
 a. 22 b. 18
 c. 17 d. 21

15. Binary Search requires
 a. sorted array b. unsorted array
 c. array sorted in ascending order d. array sorted in descending order

16. What happens when you pass an array to a method in Java?
 a. A copy of the array is passed b. A reference to the array is passed
 c. The array is converted to a list d. Only the first element is passed

17. What will arr[arr.length - 1] return?
 a. First element
 b. Middle element
 c. Last element
 d. Index out of range

18. Which keyword is used to allocate memory for an array?
 a. create b. make
 c. allocate d. new

19. What will the following code print?
```
int[] a = {10, 20, 30};
System.out.println(a[2]);
```
 a. 10 b. 20
 c. 30 d. Error

20. int a[]={1,2,3,4,5};
```
System.out.println(a[a[3]]);
```
 a. 4 b. 5
 c. 3 d. Error

SAQ. **10 x 2 = 20**

1. Find Output:
```
int[] arr = {3, 1, 4, 1, 5, 9};
int result = 0;

for (int i = 1; i < arr.length; i += 2) {
   result += arr[i] * arr[i - 1];
}
System.out.println(result);
```

2. What will be the value of arr[3] after the following loop executes?

```
int[] arr = {10, 20, 30, 40, 50};

for (int i = 0; i < arr.length - 1; i++) {
   arr[i + 1] = arr[i] - arr[i + 1];
}
```

3. Find Output:

```
int[] arr = new int[5];
arr[0] = 1;

for (int i = 1; i < arr.length; i++) {
   arr[i] = arr[i - 1] * 2;
}
System.out.println(arr[4]);
```

4.

```
int function(int[] arr) {
   int index = 0;
   for (int i = 1; i < arr.length; i++) {
      if (arr[i] > arr[index]) {
         index = i;
      }
   }
   return index;
}
```

If arr = {5, 10, 7, 10, 3}, what is the return value?

5. Find Output

```
public class ArrayTest {
   int modify(int[] arr) {
      arr[2] = arr[0] + arr[1];
      return arr[2];
   }

   public static void main(String[] args) {
      int[] a = {4, 5, 6};
      ArrayTest obj = new ArrayTest();
      int x = obj.modify(a);
      System.out.println(a[2] + " " + x);
   }
}
```

6. Find Output

```
class ArraySum {
   int sArray(int[] arr) {
      int sum = 0;
      for (int i=0; i<arr.length; i++) {
         sum += arr[i];
      }
      return sum;
   }
   public static void main(String[] args) {
      int[] a = {1, 2, 3, 4};
      ArraySum ob = new ArraySum();
      System.out.println(ob.sArray(a));
```

```
    }
}
```

7. Find Output

```
class ArrayCheck {
    boolean function(int[] arr) {
        for (int i=0; i<arr.length; i++) {
            if (arr[i] <= 0)
                return false;
        }
        return true;
    }

    public static void main(String[] args) {
        int[] a = {5, 3, 2, -1};
        ArrayCheck ob = new ArrayCheck();
        System.out.println(ob.function(a));
    }
}
```

8. Find output

```
int[] arr = {1, 2, 3, 4, 5};
for (int i = arr.length - 2; i >= 0; i--) {
    arr[i] = arr[i] + arr[i + 1];
}
System.out.println(arr[3]);
```

9. Find output

```
public class Program {
    public static void main(String[] args) {
        int[] a = {2, 3, 5, 7, 11};
        for (int i = 0; i < a.length - 1; i++) {
            a[i + 1] = a[i + 1] - a[i];
        }

        for (int i =0; i<a.length; i++) {
            System.out.print(a[i]);
        }
    }
}
```

10. Find Errors:

```
class Program{
        public static void main(String args[]){

                int a[]=new int[5];
                a={1,2,3,4,5};
                for (int i =0; i<a.length; i++) {
                  System.out.print(a[i]);
                }

        }
}
```

Programming: (Solve any 3) 10 x 3= 30

1. Initialize the array [2, 4, 6, 8, 15, 17, 20, 25, 30, 35]. Find their mean, median and mode programmatically.

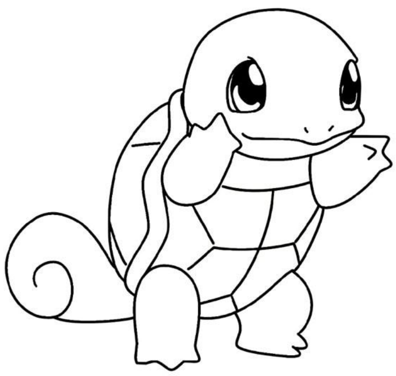
2. Design a method void Frequency(int arr[]) to count and print frequency of each single digit number in array

3. Take N elements in an array and print only prime ones. Also print the largest and smallest element

4. Write a program to take N integers in an array. Print sum of all even integers present at odd indexes and product of all odd integers at even indexes

5. Write a program to input and store roll numbers and marks in 3 subjects of n number of students in four single dimensional arrays and display the remark based on average marks as given below:

Average Marks	**Remarks**
85-100	EXCELLENT
75-84	DISTINCTION
60-74	FIRST CLASS
40-59	PASS
Less Than 40	FAIL



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